

A derived quadratic equation from a real quadratic equation with two distinct positive roots

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Abstract

Given a real quadratic equation

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

with two distinct positive roots $\alpha > \beta > 0$, this paper derives the quadratic equation whose roots are

$$\alpha + \mu, \quad \beta + \nu,$$

where $\mu, \nu \in \mathbb{R}$. The derivation uses Viète's formulae, symmetric expressions, discriminant algebra, and ordered-root reconstruction. We also discuss the transformation from statistical, data-scientific, and machine-learning viewpoints, including sensitivity analysis, uncertainty propagation, and an algorithmic implementation.

The paper ends with “The End”

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1 Problem Statement

Let

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

be a real quadratic equation with roots

$$\alpha > \beta > 0.$$

By Viète's formulae,

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

The goal is to find, in terms of only a, b, c, μ, ν , the quadratic equation with roots

$$\alpha + \mu, \quad \beta + \nu.$$

The problem is not merely a uniform translation of both roots unless $\mu = \nu$. Since the larger root α receives the shift μ and the smaller root β receives the shift ν , the answer necessarily depends on the ordered-root separation $\alpha - \beta$.

2 Preliminaries

Definition 1 (Discriminant). *For the quadratic polynomial*

$$p(x) = ax^2 + bx + c,$$

the discriminant is

$$\Delta = b^2 - 4ac.$$

Since the roots are real and distinct,

$$\Delta > 0.$$

Since the roots are positive,

$$\alpha + \beta > 0, \quad \alpha\beta > 0.$$

Equivalently,

$$-\frac{b}{a} > 0, \quad \frac{c}{a} > 0.$$

Lemma 1 (Ordered-root reconstruction). *Let $ax^2 + bx + c = 0$ have roots $\alpha > \beta$. Then*

$$\alpha = \frac{1}{2} \left(-\frac{b}{a} + \sqrt{\frac{b^2 - 4ac}{a^2}} \right), \quad \beta = \frac{1}{2} \left(-\frac{b}{a} - \sqrt{\frac{b^2 - 4ac}{a^2}} \right).$$

Proof. By Viète's formulae,

$$S = \alpha + \beta = -\frac{b}{a}, \quad P = \alpha\beta = \frac{c}{a}.$$

Also,

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = S^2 - 4P.$$

Therefore,

$$\alpha - \beta = \sqrt{S^2 - 4P} = \sqrt{\frac{b^2}{a^2} - \frac{4c}{a}} = \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

Since $\alpha > \beta$,

$$\alpha = \frac{S + (\alpha - \beta)}{2}, \quad \beta = \frac{S - (\alpha - \beta)}{2}.$$

Substituting $S = -b/a$ gives the result. □

3 Main Derivation

Let the desired roots be

$$r_1 = \alpha + \mu, \quad r_2 = \beta + \nu.$$

The desired monic quadratic equation is

$$x^2 - (r_1 + r_2)x + r_1r_2 = 0.$$

3.1 The Sum of the New Roots

The new sum is

$$r_1 + r_2 = (\alpha + \mu) + (\beta + \nu) = \alpha + \beta + \mu + \nu.$$

Using Viète's formula,

$$r_1 + r_2 = -\frac{b}{a} + \mu + \nu.$$

3.2 The Product of the New Roots

The new product is

$$r_1r_2 = (\alpha + \mu)(\beta + \nu).$$

Expanding,

$$r_1r_2 = \alpha\beta + \nu\alpha + \mu\beta + \mu\nu.$$

Now write

$$S = \alpha + \beta = -\frac{b}{a}, \quad \delta = \alpha - \beta = \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

Then

$$\alpha = \frac{S + \delta}{2}, \quad \beta = \frac{S - \delta}{2}.$$

Therefore,

$$\nu\alpha + \mu\beta = \nu\frac{S + \delta}{2} + \mu\frac{S - \delta}{2}.$$

Hence

$$\nu\alpha + \mu\beta = \frac{(\mu + \nu)S + (\nu - \mu)\delta}{2}.$$

Substituting $S = -b/a$ and

$$\delta = \sqrt{\frac{b^2 - 4ac}{a^2}},$$

we obtain

$$r_1r_2 = \frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

4 Main Theorem

Theorem 1 (Derived quadratic equation). *Let*

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

have two distinct positive real roots $\alpha > \beta > 0$. Let $\mu, \nu \in \mathbb{R}$. Then the quadratic equation whose roots are

$$\alpha + \mu, \quad \beta + \nu$$

is

$$x^2 + \left(\frac{b}{a} - \mu - \nu\right)x + \left[\frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2}\sqrt{\frac{b^2 - 4ac}{a^2}}\right] = 0.$$

Proof. The desired roots are

$$r_1 = \alpha + \mu, \quad r_2 = \beta + \nu.$$

Their sum is

$$r_1 + r_2 = -\frac{b}{a} + \mu + \nu.$$

Their product is

$$r_1 r_2 = \frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2}\sqrt{\frac{b^2 - 4ac}{a^2}}.$$

Therefore the required monic quadratic equation is

$$x^2 - (r_1 + r_2)x + r_1 r_2 = 0.$$

Substituting the above sum and product gives

$$x^2 + \left(\frac{b}{a} - \mu - \nu\right)x + \left[\frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2}\sqrt{\frac{b^2 - 4ac}{a^2}}\right] = 0.$$

□

5 Equivalent Non-Monic Form

Multiplying the monic equation by any nonzero scalar produces an equivalent quadratic equation. For example, multiplying by a gives

$$ax^2 + (b - a\mu - a\nu)x + \left[c + a\mu\nu - \frac{b(\mu + \nu)}{2} + \frac{a(\nu - \mu)}{2}\sqrt{\frac{b^2 - 4ac}{a^2}}\right] = 0.$$

Because

$$a\sqrt{\frac{b^2 - 4ac}{a^2}} = \text{sgn}(a)\sqrt{b^2 - 4ac},$$

this may also be written as

$$ax^2 + (b - a\mu - a\nu)x + \left[c + a\mu\nu - \frac{b(\mu + \nu)}{2} + \frac{\text{sgn}(a)(\nu - \mu)}{2}\sqrt{b^2 - 4ac}\right] = 0.$$

6 Special Cases

Case	Roots	Resulting equation
$\mu = \nu = 0$	α, β	$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$
$\mu = \nu = \lambda$	$\alpha + \lambda, \beta + \lambda$	$x^2 + \left(\frac{b}{a} - 2\lambda\right)x + \left(\frac{c}{a} - \frac{b\lambda}{a} + \lambda^2\right) = 0$
$\nu = \mu$	equal shift	No discriminant-asymmetry term
$\nu - \mu = \alpha - \beta$	$\alpha + \mu = \beta + \nu$	Derived quadratic has repeated root

Table 1: Important special cases of the derived quadratic equation.

Corollary 1 (Uniform translation). *If $\mu = \nu = \lambda$, then the required equation is obtained by substituting $x - \lambda$ into the original polynomial:*

$$a(x - \lambda)^2 + b(x - \lambda) + c = 0.$$

After division by a , this becomes

$$x^2 + \left(\frac{b}{a} - 2\lambda\right)x + \left(\frac{c}{a} - \frac{b\lambda}{a} + \lambda^2\right) = 0.$$

Proof. If $\mu = \nu = \lambda$, both roots are shifted by the same amount. Therefore, if y is a root of the original equation, then $x = y + \lambda$, or $y = x - \lambda$. Substituting $x - \lambda$ into

$$ay^2 + by + c = 0$$

gives

$$a(x - \lambda)^2 + b(x - \lambda) + c = 0.$$

□

7 Symmetric and Antisymmetric Decomposition

The transformation naturally decomposes into a symmetric part and an antisymmetric part.

Let

$$m = \frac{\mu + \nu}{2}, \quad d = \frac{\mu - \nu}{2}.$$

Then

$$\mu = m + d, \quad \nu = m - d.$$

The new roots are

$$\alpha + m + d, \quad \beta + m - d.$$

The common component m shifts both roots equally, while d widens or narrows the separation asymmetrically.

The new sum is

$$\alpha + \beta + 2m = -\frac{b}{a} + 2m.$$

The new product is

$$(\alpha + m + d)(\beta + m - d).$$

Expanding,

$$(\alpha + m + d)(\beta + m - d) = \alpha\beta + m(\alpha + \beta) + m^2 + d(\beta - \alpha) - d^2.$$

Since

$$\beta - \alpha = -\sqrt{\frac{b^2 - 4ac}{a^2}},$$

we get

$$r_1 r_2 = \frac{c}{a} - \frac{bm}{a} + m^2 - d\sqrt{\frac{b^2 - 4ac}{a^2}} - d^2.$$

This is equivalent to the theorem after substituting

$$m = \frac{\mu + \nu}{2}, \quad d = \frac{\mu - \nu}{2}.$$

8 Algorithmic Form

The derivation yields a direct computational procedure.

Step	Operation
1	Input a, b, c, μ, ν , with $a \neq 0$.
2	Compute the discriminant $\Delta = b^2 - 4ac$.
3	Check that $\Delta > 0$. For the stated problem, this must hold.
4	Compute the ordered-root gap $\delta = \sqrt{\frac{\Delta}{a^2}}.$
5	Compute the new sum $T = -\frac{b}{a} + \mu + \nu.$
6	Compute the new product $U = \frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2}\delta.$
7	Output $x^2 - Tx + U = 0.$

9 Pseudocode

Algorithm 1: Derived quadratic from asymmetric root shifts

1. Receive input a, b, c, μ, ν .
2. Compute $\Delta \leftarrow b^2 - 4ac$.
3. If $a = 0$, stop: the original equation is not quadratic.
4. If $\Delta \leq 0$, stop: the original equation does not have two distinct real roots.
5. Compute

$$\delta \leftarrow \sqrt{\Delta/a^2}.$$

6. Compute

$$T \leftarrow -b/a + \mu + \nu.$$

7. Compute

$$U \leftarrow c/a + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2}\delta.$$

8. Return the quadratic equation

$$x^2 - Tx + U = 0.$$

10 Computational Complexity

The computation uses only a fixed number of arithmetic operations and one square-root operation. Therefore the arithmetic complexity is constant.

Quantity	Complexity
Discriminant computation	$O(1)$
Root-gap computation	$O(1)$
New sum computation	$O(1)$
New product computation	$O(1)$
Total complexity	$O(1)$

Table 2: Computational complexity of the derived quadratic construction.

11 Numerical Stability

The formula avoids explicitly computing both roots α and β . Instead, it uses the symmetric quantities

$$\alpha + \beta, \quad \alpha\beta, \quad \alpha - \beta.$$

This is usually preferable because it reduces unnecessary root reconstruction.

However, the expression

$$\sqrt{b^2 - 4ac}$$

can suffer from floating-point cancellation if b^2 and $4ac$ are close. This corresponds to the case where the two roots are nearly equal. In such cases, the original quadratic is close to having a repeated root, and the ordered-root gap

$$\delta = \alpha - \beta$$

is small.

Remark 1. *The asymmetric term*

$$\frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}$$

is the only part of the derived product that depends on the ordering $\alpha > \beta$. If $\mu = \nu$, this term vanishes.

12 Statistical Interpretation

Suppose the coefficients a, b, c are estimated from data. For example, one might fit a quadratic model

$$y_i = ax_i^2 + bx_i + c + \varepsilon_i,$$

where

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

Then the estimated coefficient vector is

$$\hat{\theta} = (\hat{a}, \hat{b}, \hat{c})^\top.$$

The derived quadratic coefficients become nonlinear functions of $\hat{\theta}$.

Define

$$B(a, b, c, \mu, \nu) = \frac{b}{a} - \mu - \nu$$

and

$$C(a, b, c, \mu, \nu) = \frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

Then the derived quadratic is

$$x^2 + Bx + C = 0.$$

If

$$\hat{\theta} \approx N(\theta, \Sigma),$$

then the delta method gives

$$\begin{pmatrix} \hat{B} \\ \hat{C} \end{pmatrix} \approx N \left(\begin{pmatrix} B \\ C \end{pmatrix}, J\Sigma J^\top \right),$$

where

$$J = \begin{pmatrix} \nabla_\theta B \\ \nabla_\theta C \end{pmatrix}.$$

This gives approximate standard errors and confidence intervals for the derived quadratic coefficients.

13 Data-Science Viewpoint

In a data-science pipeline, the map

$$(a, b, c, \mu, \nu) \mapsto (1, B, C)$$

can be treated as a deterministic feature transformation.

Input feature	Mathematical role	Output contribution
a	leading coefficient	affects normalization and discriminant
b	linear coefficient	affects sum and discriminant
c	constant coefficient	affects product and discriminant
μ	shift of larger root	asymmetric root transformation
ν	shift of smaller root	asymmetric root transformation
$\Delta = b^2 - 4ac$	root separability	determines ordered-root gap

Table 3: Feature interpretation of the derived quadratic transformation.

The transformation is interpretable because each component has a direct algebraic meaning:

$$B = \frac{b}{a} - \mu - \nu$$

controls the negative sum of the new roots, while C controls their product.

14 Machine-Learning Interpretation

A machine-learning model could be trained to approximate the map

$$F : \mathbb{R}^5 \rightarrow \mathbb{R}^2, \quad F(a, b, c, \mu, \nu) = (B, C),$$

where

$$B = \frac{b}{a} - \mu - \nu$$

and

$$C = \frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

However, since the exact symbolic form is known, a purely black-box model is unnecessary. A better approach is a physics-informed or structure-informed model that embeds the exact formula as a differentiable layer.

For example, let a neural network produce estimates

$$(\hat{a}, \hat{b}, \hat{c}, \hat{\mu}, \hat{\nu})$$

from raw data. A deterministic algebraic layer can then compute

$$\hat{B} = \frac{\hat{b}}{\hat{a}} - \hat{\mu} - \hat{\nu}$$

and

$$\hat{C} = \frac{\hat{c}}{\hat{a}} + \hat{\mu}\hat{\nu} - \frac{\hat{b}(\hat{\mu} + \hat{\nu})}{2\hat{a}} + \frac{\hat{\nu} - \hat{\mu}}{2} \sqrt{\frac{\hat{b}^2 - 4\hat{a}\hat{c}}{\hat{a}^2}}.$$

This gives an interpretable hybrid model.

A possible supervised loss is

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \left[(\hat{B}_i - B_i)^2 + (\hat{C}_i - C_i)^2 \right].$$

If root predictions are also available, one may add

$$\mathcal{L}_{\text{root}} = \frac{1}{n} \sum_{i=1}^n \left[(\hat{r}_{1i} - r_{1i})^2 + (\hat{r}_{2i} - r_{2i})^2 \right].$$

15 Sensitivity Analysis

The coefficient of x in the derived quadratic is

$$B = \frac{b}{a} - \mu - \nu.$$

Its partial derivatives are

$$\frac{\partial B}{\partial a} = -\frac{b}{a^2}, \quad \frac{\partial B}{\partial b} = \frac{1}{a}, \quad \frac{\partial B}{\partial c} = 0,$$

and

$$\frac{\partial B}{\partial \mu} = -1, \quad \frac{\partial B}{\partial \nu} = -1.$$

The constant coefficient is

$$C = \frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

Let

$$G = \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

Then

$$C = \frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2} G.$$

The sensitivities with respect to μ and ν are especially transparent:

$$\frac{\partial C}{\partial \mu} = \nu - \frac{b}{2a} - \frac{1}{2} G,$$

and

$$\frac{\partial C}{\partial \nu} = \mu - \frac{b}{2a} + \frac{1}{2} G.$$

Using

$$\alpha = \frac{1}{2} \left(-\frac{b}{a} + G \right), \quad \beta = \frac{1}{2} \left(-\frac{b}{a} - G \right),$$

these become

$$\frac{\partial C}{\partial \mu} = \beta + \nu, \quad \frac{\partial C}{\partial \nu} = \alpha + \mu.$$

This agrees with the product rule:

$$C = (\alpha + \mu)(\beta + \nu).$$

16 Monte Carlo Estimation

If the coefficients are random or estimated with uncertainty, Monte Carlo simulation can approximate the distribution of the derived coefficients.

Algorithm 2: Monte Carlo uncertainty propagation

1. Specify a distribution for (a, b, c, μ, ν) .

2. For $k = 1, \dots, N$, sample

$$(a_k, b_k, c_k, \mu_k, \nu_k).$$

3. Retain only samples satisfying

$$a_k \neq 0, \quad b_k^2 - 4a_k c_k > 0.$$

4. Compute

$$B_k = \frac{b_k}{a_k} - \mu_k - \nu_k.$$

5. Compute

$$C_k = \frac{c_k}{a_k} + \mu_k \nu_k - \frac{b_k(\mu_k + \nu_k)}{2a_k} + \frac{\nu_k - \mu_k}{2} \sqrt{\frac{b_k^2 - 4a_k c_k}{a_k^2}}.$$

6. Estimate

$$\mathbb{E}[B], \quad \mathbb{E}[C], \quad \text{Var}(B), \quad \text{Var}(C), \quad \text{Cov}(B, C).$$

17 Vector Graphics

Figure 1 illustrates the ordered-root transformation.

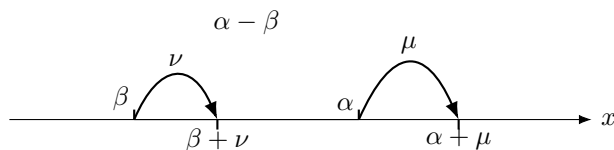


Figure 1: Ordered roots $\alpha > \beta$ and their asymmetric shifts.

Figure 2 gives a dependency diagram for the construction.

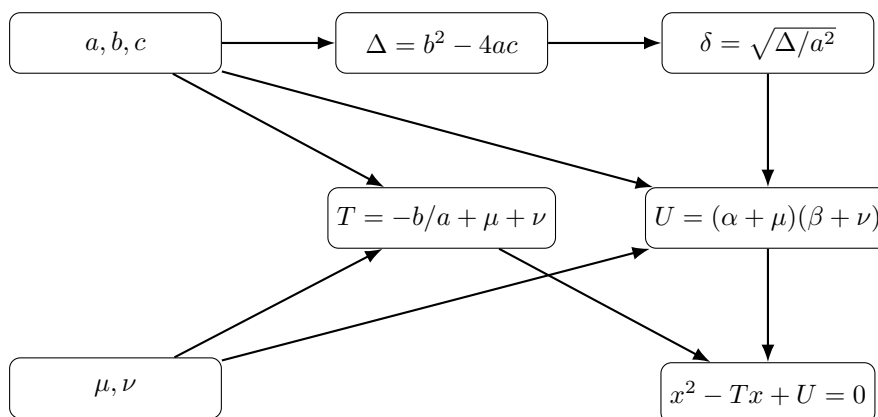


Figure 2: Dependency graph for the derived quadratic coefficients.

18 Worked Example

Consider

$$x^2 - 5x + 6 = 0.$$

Here

$$a = 1, \quad b = -5, \quad c = 6.$$

The roots are

$$\alpha = 3, \quad \beta = 2.$$

Let

$$\mu = 4, \quad \nu = 1.$$

Then the desired roots are

$$\alpha + \mu = 7, \quad \beta + \nu = 3.$$

The derived quadratic should therefore be

$$(x - 7)(x - 3) = x^2 - 10x + 21 = 0.$$

Using the formula,

$$\Delta = b^2 - 4ac = 25 - 24 = 1,$$

so

$$\sqrt{\frac{\Delta}{a^2}} = 1.$$

The coefficient of x is

$$\frac{b}{a} - \mu - \nu = -5 - 4 - 1 = -10.$$

The constant term is

$$\frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

Substituting,

$$6 + 4 - \frac{(-5)(5)}{2} + \frac{1 - 4}{2} = 10 + \frac{25}{2} - \frac{3}{2} = 10 + 11 = 21.$$

Therefore,

$$x^2 - 10x + 21 = 0.$$

19 Conclusion

The derived quadratic equation with roots

$$\alpha + \mu, \quad \beta + \nu$$

is

$$x^2 + \left(\frac{b}{a} - \mu - \nu\right)x + \left[\frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}\right] = 0.$$

The formula is symmetric in the original roots except for the necessary ordered-root correction

$$\frac{\nu - \mu}{2} \sqrt{\frac{b^2 - 4ac}{a^2}}.$$

This correction appears because μ is attached specifically to the larger root α , while ν is attached specifically to the smaller root β .

References

- [1] G. H. Hardy, *A Course of Pure Mathematics*, Cambridge University Press.
- [2] Serge Lang, *Algebra*, Springer.
- [3] Nicholas J. Higham, *Accuracy and Stability of Numerical Algorithms*, Society for Industrial and Applied Mathematics.
- [4] George Casella and Roger L. Berger, *Statistical Inference*, Duxbury.
- [5] Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
- [6] Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *The Elements of Statistical Learning*, Springer.
- [7] Donald E. Knuth, *The Art of Computer Programming*, Addison-Wesley.

Glossary

Quadratic equation An equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$.

Root A value $x = r$ satisfying $ar^2 + br + c = 0$.

Viète's formulae Relations between the coefficients and roots of a polynomial. For a quadratic,

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

Discriminant The quantity

$$\Delta = b^2 - 4ac.$$

It determines the nature of the roots of a quadratic equation.

Ordered roots A labeling of the two real roots such that $\alpha > \beta$.

Symmetric expression An expression in α, β unchanged under the exchange $\alpha \leftrightarrow \beta$, such as $\alpha + \beta$ or $\alpha\beta$.

Antisymmetric expression An expression that changes sign under the exchange $\alpha \leftrightarrow \beta$, such as $\alpha - \beta$.

Delta method A statistical approximation technique used to estimate the distribution of a nonlinear function of an asymptotically normal estimator.

Monte Carlo simulation A computational method that estimates distributions or expectations by repeated random sampling.

Structure-informed model A machine-learning model that embeds known mathematical structure instead of learning everything as an unconstrained black box.

The End